Icet Ethan
Problem Set 7

Problem 1

We have $H(X,Y,Z) \geq H(X,Z)$ by Property 3: increase of entropy by "adding" new randomness

Property 2: entropy chain rule
$$\begin{cases} H(X,Y,Z) = H(Z) + H(X,Y|Z) \\ H(X,Z) = H(Z) + H(X|Z) \end{cases}$$

$$\Rightarrow H(Z) + H(X,Y|Z) \geq H(Z) + H(X|Z)$$
$$\Rightarrow H(X,Y|Z) \geq H(X|Z)$$

Problem 2

X	Y	$P_{XY}(x,y)$
0	0	1/8
0	1	1/4
1	0	1/8
1	1	1/4
2	0	0
2	1	1/4

$P_X(x) = \sum_{y \in \mathcal{Y}} P_{XY}(x,y)$

$P_Y(y) = \sum_{x \in \mathcal{X}} P_{XY}(x,y)$

$p(X=0) = p(X=0, Y=0) + p(X=0, Y=1) = 1/8 + 1/4 = \frac{3}{8}$

$p(X=1) = 1/8 + 1/4 = \frac{3}{8}$

$p(X=2) = 0 + 1/4 = 1/4$

$p(Y=0) = \frac{1}{8} + \frac{1}{8} + 0 = \frac{1}{4}$

$p(Y=1) = 1/4 + 1/4 + 1/4 = \frac{3}{4}$

$$I(X;Y) = E_{p(x,y)}\left[\log \frac{p(x,y)}{p(x)p(y)}\right]$$

$$= \sum_{x \in \mathcal{X}, y \in \mathcal{Y}} p(x,y) \log \frac{p(x,y)}{p(x)p(y)} = p(X=0,Y=0)\log_2 \frac{p(X=0,Y=0)}{p(X=0)p(Y=0)} + p(X=0,Y=1)\log_2 \frac{p(X=0,Y=1)}{p(X=0)p(Y=1)}$$
$$+ p(X=1,Y=0)\log_2 \frac{p(X=1,Y=0)}{p(X=1)p(Y=0)} + p(X=1,Y=1)\log_2 \frac{p(X=1,Y=1)}{p(X=1)p(Y=1)}$$
$$+ p(X=2,Y=0)\log_2 \frac{p(X=2,Y=0)}{p(X=2)p(Y=0)} + p(X=2,Y=1)\log_2 \frac{p(X=2,Y=1)}{p(X=2)p(Y=1)}$$

$$= \frac{1}{8}\log_2 \frac{1/8}{3/8 \cdot 1/4} + \frac{1}{4}\log_2 \frac{1/4}{3/8 \cdot 3/4}$$
$$+ \frac{1}{8}\log_2 \frac{1/8}{3/8 \cdot 1/4} + \frac{1}{4}\log_2 \frac{1/4}{3/8 \cdot 3/4}$$
$$+ 0 + \frac{1}{4}\log_2 \frac{1/4}{1/4 \cdot 3/4}$$

$$\simeq 0.1226 \text{ (bits)}$$

a) $H(X) = -E_{p(x)}[\log p(x)] = -\sum_{x \in \mathcal{X}} p(x) \log p(x)$

$= -\left(p(x=0)\log_2 p(x=0) + p(x=1)\log_2 p(x=1) + p(x=2)\log_2 p(x=2)\right)$

$= -\left(\frac{3}{8}\log_2 \frac{3}{8} + \frac{3}{8}\log_2 \frac{3}{8} + \frac{1}{4}\log_2 \frac{1}{4}\right)$

≈ 1.5613 (bits)

$H(X|Y) = -E_{p(x,y)}[\log p(x|y)] = \sum_{x \in \mathcal{X}, y \in \mathcal{Y}} p(x,y) \log \frac{p(y)}{p(x,y)}$

$= \frac{1}{8}\log_2 \frac{1/4}{1/8} + \frac{1}{4}\log_2 \frac{3/4}{1/4} +$

$\frac{1}{8}\log_2 \frac{1/4}{1/8} + \frac{1}{4}\log_2 \frac{3/4}{1/4} +$

$0 + \frac{1}{4}\log_2 \frac{3/4}{1/4}$

≈ 1.4387 (bits)

$I(X;Y) = E_{p(x,y)}\left[\log \frac{p(x,y)}{p(x)p(y)}\right] = E_{p(x,y)}\left[\log \left(\frac{p(x,y)}{p(y)} \cdot \frac{1}{p(x)}\right)\right]$

$= E_{p(x,y)}\left[\log\left(\frac{p(x,y)}{p(y)}\right) - \log p(x)\right]$

$= E_{p(x,y)}[\log(p(x|y))] - E_{p(x,y)}[\log p(x)]$

$= -H(X|Y) + H(X)$

$H(X) - H(X|Y) = I(X,Y)$

$\sim 1.5613 - 1.4387 = 0.1226$

b) $H(Y) = -\sum p(y) \log p(y) = -\left(\frac{1}{4}\log_2 \frac{1}{4} + \frac{3}{4}\log_2 \frac{3}{4}\right)$

≈ 0.8113 (bits)

$H(Y|X) = -\sum_{x \in \mathcal{X}, y \in \mathcal{Y}} p(x,y) \log \frac{p(x)}{p(x,y)} = \frac{1}{8}\log_2 \frac{3/8}{1/8} + \frac{1}{4}\log_2 \frac{3/8}{1/4}$

$+ \frac{1}{8}\log_2 \frac{3/8}{1/8} + \frac{1}{4}\log_2 \frac{3/8}{1/4}$

$+ 0 + \frac{1}{4}\log_2 \frac{1/4}{1/4}$

≈ 0.6887 (bits)

$I(X;Y) = E_{p(x,y)}\left[\log \frac{p(x,y)}{p(x)p(y)}\right]$

$= E_{p(x,y)}\left[\log \frac{p(x,y)}{p(x)} \cdot \frac{1}{p(y)}\right]$

$= E_{p(x,y)}\left[\log \frac{p(x,y)}{p(x)} - \log p(y)\right]$

$= E_{p(x,y)}\left[\log \frac{p(x,y)}{p(x)}\right] - E_{p(x,y)}[\log p(y)] = -H(Y|X) + H(Y)$

$H(Y) - H(Y|X) = I(X,Y)$

$\sim 0.8113 - 0.6887 = 0.1226$

[Python] Problem 3 $\quad p(x) = 0.125,\ x \in \{1, 2, \dots, 8\}$

Results: a) $D_{KL}(p\|q) \approx 0.3131 \quad (D_{KL}(q\|p) \approx 0.2704)$

b) $D_{KL}(p\|q) = +\infty \quad (D_{KL}(q\|p) \approx 1.153)$

Problem 4 $\quad D(P_x \| Q_x) = d(p,q) = p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q}$

i $q = p \Rightarrow d(p,q) = p \log \frac{p}{p} + (1-p) \log \frac{1-p}{1-p}$

$= p \log 1 + (1-p) \log 1$

$= 0 + 0 = 0$

ii $p = 0 \Rightarrow d(p,q) = \underbrace{0 \cdot \log \frac{0}{q}}_{\text{convention}} + (1-0) \log \frac{1-0}{1-q}$

$= 0 + 1 \log \frac{1}{1-q}$

$= -\log(1-q)$

iii $p = 1 \Rightarrow d(p,q) = 1 \log \frac{1}{q} + \underbrace{(1-1) \log \frac{1-1}{1-q}}_{\text{Convention}}$

$= \log q^{-1} + 0$

$= -\log q$

[Python]

Problem 5

$$H(p(x), q(x)) = \sum_{x \in \mathcal{X}} p(x) \log\left(\frac{1}{q(x)}\right) = \sum_{x \in \mathcal{X}} p(x) \log\left(\frac{1}{q(x)} \cdot \frac{p(x)}{p(x)}\right)$$

$$= \sum_{x \in \mathcal{X}} p(x) \left[\log\left(\frac{p(x)}{q(x)}\right) + \log\left(\frac{1}{p(x)}\right)\right] = -\sum_{x \in \mathcal{X}} p(x) \log p(x) + \sum_{x \in \mathcal{X}} p(x) \log \frac{p(x)}{q(x)}$$

$$= H(p(x)) + D_{KL}(p\|q)$$

$$\arg\min_{q(x)} H(p(x), q(x)) = \arg\min_{q(x)} \underbrace{H(p(x))}_{\text{Constant, does it depend on } q(x)} + D_{KL}(p(x) \| q(x))$$

$$\Rightarrow \arg\min_{q(x)} H(p(x), q(x)) = \arg\min_{q(x)} D_{KL}(p(x) \| q(x))$$